

Exercise – Understanding hypothesis testing using simulation

1. Aim of the exercise

The aim of this exercise is to explain hypothesis testing using the t test through simulation. Without simulation, the logic underlying hypothesis testing can be difficult to grasp, which is why students often end up memorizing the steps of the procedure rather than developing an intuitive understanding of how it works.

2. Define the t value and the degrees of freedom

To illustrate the logic of hypothesis testing, suppose that using some sample data, a t test yields the test statistic

$$t = \frac{\hat{\beta}_k - \beta_k^0}{SE(\hat{\beta}_k)} = 1.8$$

and that the appropriate number of degrees of freedom is 20. These values are assumed for the purpose of this exercise. We would like to determine whether this hypothetical test statistic provides sufficient evidence against the null hypothesis at the 0.05 significance level. Specifically, we wish to test the null hypothesis

$$H_0 : \beta_k \leq \beta_k^0$$

against the alternative hypothesis

$$H_1 : \beta_k > \beta_k^0.$$

This is a right-tailed test for a regression coefficient.

```
8 %% 2. Define t value and degrees of freedom
9
10 % 2.1. Define t value
11 tValue = 1.8;
12
13 % 2.2. Define degrees of freedom
14 nu = 20;
```

3. Generate Student's t distributed values

We begin by generating a large number of random draws, in particular 20,000, from the Student's t distribution with 20 degrees of freedom and plotting the resulting frequency distribution. The draws are generated using the stochastic representation introduced in earlier exercises on the t -distribution.

It is important to keep in mind that these values are entirely hypothetical. They are not obtained from real data, nor do they correspond to the outcome of an actual experiment. Instead, each draw represents a possible value of the test statistic that could occur purely by random sampling variation when the null hypothesis is true.

By generating thousands of such draws, we obtain an approximation to the sampling distribution of the test statistic under the assumption that the null hypothesis is true, commonly referred to as the *null distribution*.

```

16 %% 3. Generate Student's t distributed values
17
18 % 3.1. Number of random draws
19 N = 20000;
20
21 % 3.2. Define histogram bin edges
22 binEdges = -6:0.01:6;
23
24 % 3.3. Construct t-distributed random variables
25 draws = StochasticRepresentation(N,nu);
26
27 % 3.4. Create histogram
28 figure;
29 set(gcf,'Position',[100 100 1000 1000]);
30 histogram(draws,...
31     'BinEdges',binEdges,...
32     'FaceColor','#b3b3b3',...
33     'EdgeAlpha',0,...
34     'DisplayName','All simulated values');
35 title(['Fig. 1. Histogram of simulated t-distributed draws, N = ', ...
36     num2str(N), ', \nu = ', num2str(nu)]);
37 xlabel('t');
38 ylabel('Frequency');
39 legend('show');

```

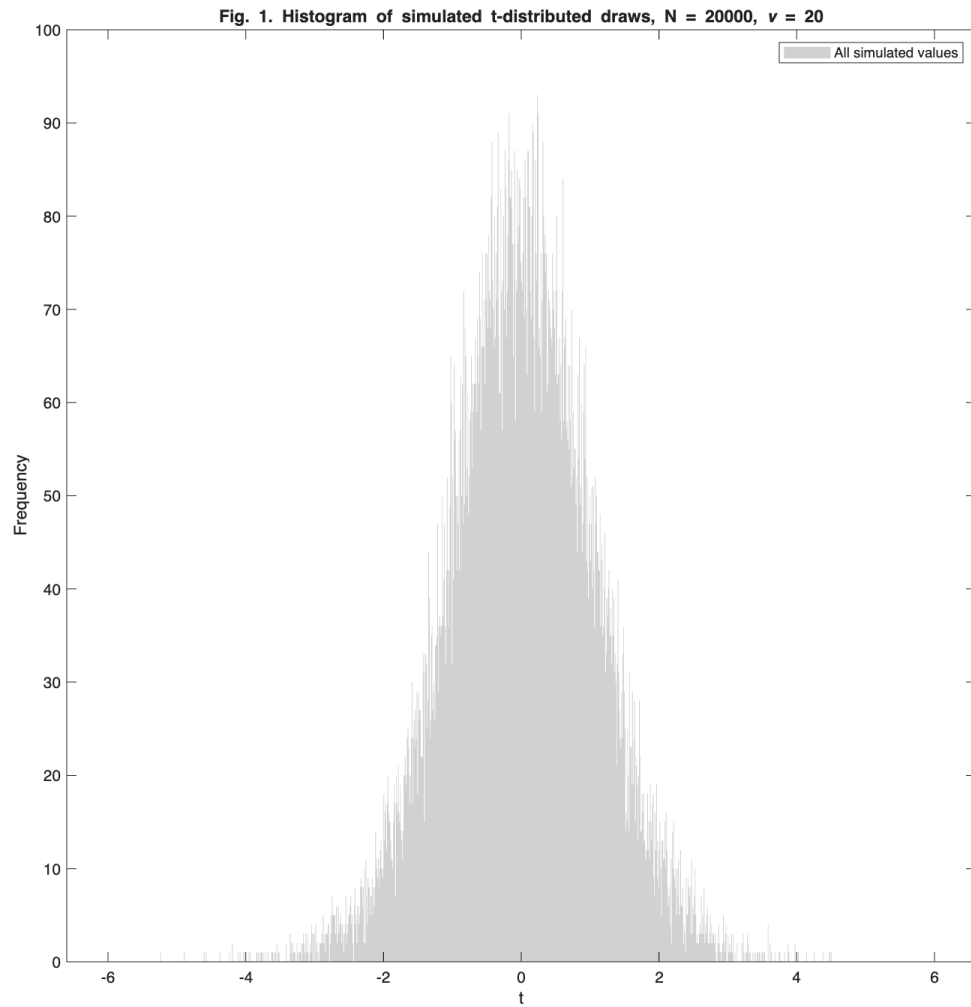


Figure 1: Approximation to the null distribution obtained from 20,000 simulated draws from a Student's t distribution with 20 degrees of freedom

4. Add the test statistic to the plot

In the frequency distribution of the simulated draws, we mark the observed test statistic from our example, which is 1.8. The vertical line indicates this value. It is important to notice that the histogram represents hypothetical values, whereas the vertical line represents the value observed in the sample. The histogram is not constructed from repeated samples of actual data. Instead, it is generated by simulation from the null distribution of the test statistic. Each simulated draw represents a possible value that the test statistic could take if the null hypothesis were true.

The observed test statistic is

$$t = 1.8.$$

Under the null hypothesis, this statistic follows a t -distribution with 20 degrees of freedom.

To determine whether the observed test statistic provides sufficient evidence against the null hypothesis, we compare it with an appropriate threshold value from the null distribution. For a right-tailed test at the 5% significance level, this threshold value is

$$t_{0.95,20} = 1.725.$$

The notation $t_{0.95,20}$ denotes the 95th percentile of the t -distribution with 20 degrees of freedom. Consequently, 95% of the distribution lies to the left of 1.725 and 5% lies to the right. The connection between this 5% right-tail area and hypothesis testing will become clearer in the next section, where we calculate the proportion of the null distribution lying to the right of the observed test statistic.

Because

$$1.8 > 1.725,$$

the observed test statistic exceeds the threshold value and therefore falls in the rejection region. We consequently reject the null hypothesis

$$H_0 : \beta_k \leq \beta_k^0$$

in favor of the alternative

$$H_1 : \beta_k > \beta_k^0.$$

Equivalently, the estimated coefficient lies 1.8 estimated standard errors above the hypothesized value β_k^0 :

$$\hat{\beta}_k - \beta_k^0 = 1.8 SE(\hat{\beta}_k).$$

Since

$$1.8 SE(\hat{\beta}_k) > 1.725 SE(\hat{\beta}_k),$$

the estimated coefficient lies farther above the hypothesized value than is required for rejection. Therefore, the test statistic falls in the rejection region, and we reject the null hypothesis in favor of the alternative.

At first glance, this logic may seem unusual. The histogram consists of hypothetical t values generated under the assumption that the null hypothesis is true, whereas the observed test statistic is computed from the actual sample data. In hypothesis testing, the idea is to ask whether the observed test statistic looks typical relative to this hypothetical distribution. If the observed test value is close to the mean of the most likely values in the center of the null

distribution, we say that the test value is typical and provides evidence in favor of the null. If it is farther away from the mean, we say that the test value is not typical of the null and provides evidence against the null.

A common source of confusion stems from the existence of two possibilities. First, if the value of the test statistic we observe is not typical, it could simply be that we observed a rare realization and the null hypothesis is still correct. Second, the null hypothesis could be false and we observed a value that is not surprising under the true distribution. This confusion becomes even stronger when one realizes that we do not observe the true data-generating distribution but instead work with the hypothetical null distribution. That is, we can never be certain whether the observed test statistic is typical of the true distribution because we observe only a single realization.

This is precisely why hypothesis testing requires a decision rule. From a single realization, we cannot determine whether an unusual observation is a rare realization under a true null hypothesis or evidence against the null hypothesis itself. Classical hypothesis testing resolves this ambiguity by taking the position that observations which are not typical of the null distribution should be interpreted as evidence against the null hypothesis rather than as rare realizations under a true null hypothesis. The significance level determines how atypical an observation must be before we reject the null hypothesis.

```
41 %% 4. Add the test statistic to the plot
42
43 % Plot histogram and test statistic
44 figure;
45 set(gcf,'Position',[100 100 1000 1000]);
46 histogram(draws,...
47     'BinEdges',binEdges,...
48     'FaceColor','#b3b3b3',...
49     'EdgeAlpha',0,...
50     'DisplayName','All simulated values');
51 hold on
52 xline(tValue,...
53     'Color','b',...
54     'DisplayName','Test statistic');
55 hold off
56 title(['Fig. 2. Histogram of simulated t-distributed draws, N = ', ...
57     num2str(N), ', \nu = ',num2str(nu)]);
58 xlabel('t');
59 ylabel('Frequency');
60 legend('show');
```

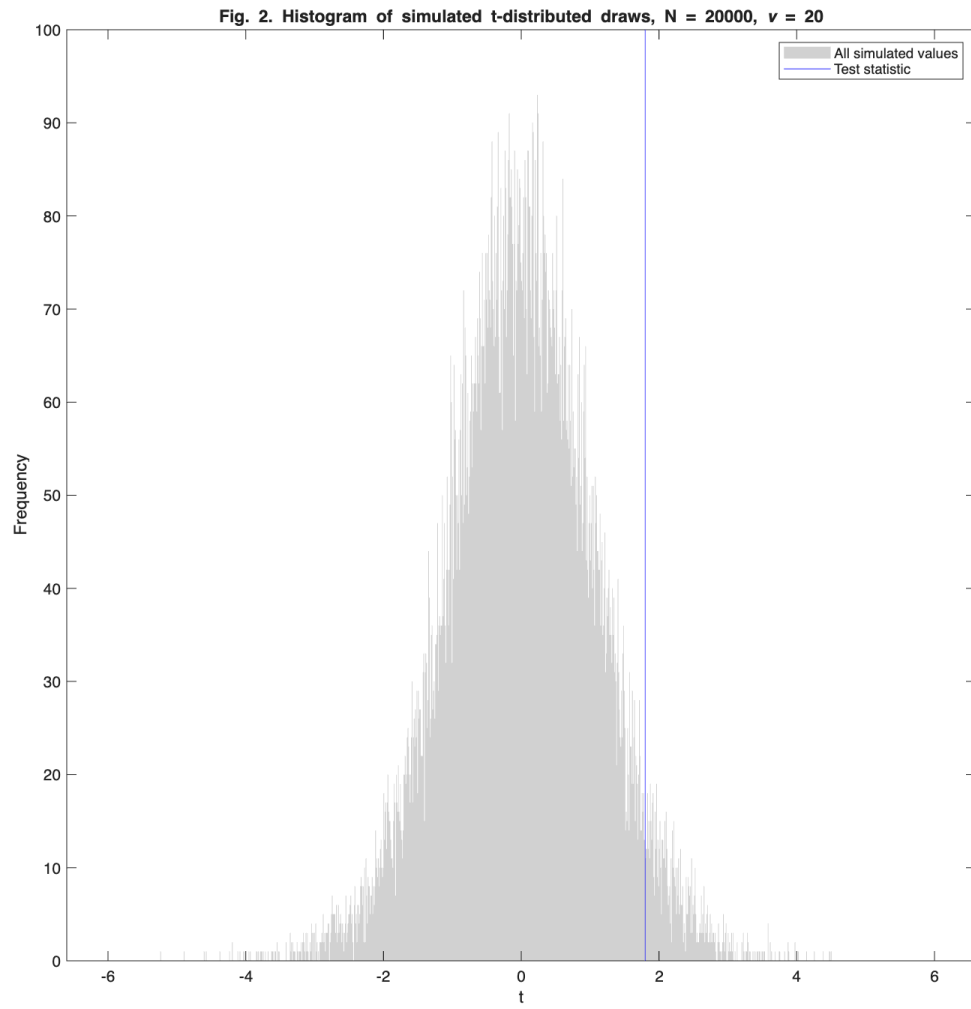


Figure 2: Approximation to the null distribution and observed test statistic

5. Highlight values more extreme than the observed test statistic

We define a variable named `extremeValues` containing all simulated draws from the null distribution that exceed the observed test statistic, 1.8. These draws are highlighted in the frequency distribution. The variable `simulatedPValue` computes the proportion of simulated draws that are at least as extreme as the observed test statistic. We obtain a value of 0.0440, indicating that approximately 4.40% of the simulated draws exceed the observed value. This simulated proportion provides an estimate of the right-tail probability, or equivalently, the one-sided p -value. Because this estimate is based on random simulation, its exact value will vary slightly from run to run. Increasing the number of simulated draws reduces this simulation error and yields a more precise estimate of the true p -value.

Since

$$4.40\% < 5.00\%,$$

the simulated p -value is smaller than the chosen significance level. We therefore reject the null hypothesis. This is the same conclusion obtained earlier using the critical-value approach.

```
62 %% 5. Highlight values more extreme than the observed test statistic
63
64 % 5.1. Select values exceeding the test statistic
65 extremeValues = draws(draws > tValue);
66
67 % 5.2. Plot histogram and Highlight values exceeding the test statistic
68 figure;
69 set(gcf,'Position',[100 100 1000 1000]);
70 histogram(draws,...
71     'BinEdges',binEdges,...
72     'FaceColor','#b3b3b3',...
73     'EdgeAlpha',0,...
74     'DisplayName','All simulated values');
75 hold on
76 histogram(extremeValues,...
77     'BinEdges',binEdges,...
78     'FaceColor','blue',...
79     'EdgeAlpha',0,...
80     'DisplayName','Values exceeding the test statistic');
81 xline(tValue,...
82     'Color','b',...
83     'DisplayName','Test statistic');
84 hold off
85 title(['Fig. 3. Simulated values exceeding the test statistic, N = ', ...
86     num2str(N), ', \nu = ',num2str(nu)]);
87 xlabel('t');
88 ylabel('Frequency');
89 legend('show');
90
91 % 5.3. Estimate the right-tail probability (simulated p-value)
92 simulatedPValue = numel(extremeValues)/numel(draws);
```

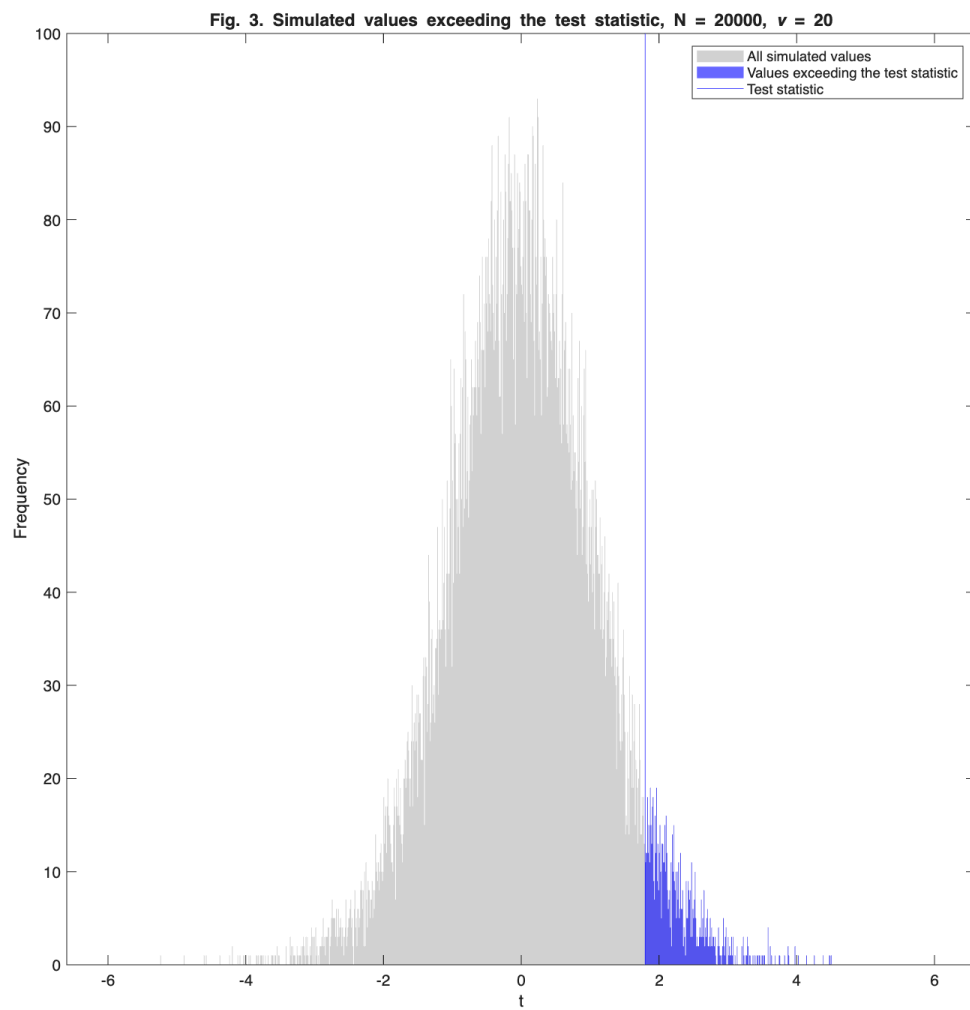


Figure 3: Approximation to the null distribution with values exceeding the observed test statistic shaded

6. Theoretical p -value calculation

In the previous section, we approximated the p -value by simulation. In practice, we usually calculate the p -value directly from the limiting distribution of the test statistic under the null hypothesis rather than approximating it by simulation. Let $X \sim t_{20}$ denote a random variable following a t -distribution with 20 degrees of freedom. Since the observed test statistic is $t = 1.8$, the corresponding one-sided p -value is

$$P(X \geq 1.8).$$

To calculate this probability, we use the cumulative distribution function, defined as

$$F_X(x) = P(X \leq x).$$

Since

$$P(X \geq 1.8) = 1 - F_X(1.8),$$

the p -value can be computed in MATLAB using the built-in `cdf` function. In this example, the theoretical p -value is 0.0435, which is very close to the simulated p -value obtained in the previous section. The small difference arises because the simulated p -value is based on a finite number of random draws, whereas the theoretical p -value is calculated directly from the limiting distribution.

```
94 %% 6. Theoretical p-value calculation
95
96 % 6.1. Define the degrees of freedom
97 nu = 20;
98
99 % 6.2. Define evaluation points
100 xValues = -6:0.1:6;
101
102 % 6.3. Evaluate the PDF of the t-distribution
103 pdfValues = PDF(xValues,nu);
104
105 % 6.4. Plot the PDF and shade the p-value area
106 figure;
107 set(gcf,'Position',[100 100 1000 1000]);
108 plot(xValues,pdfValues, ...
109      'Color','k', ...
110      'DisplayName','PDF');
111 hold on
112 xline(tValue,...
113      'Color','b', ...
114      'DisplayName','Test statistic');
115 tailRange = tValue:0.1:6;
116 tailDensity = pdf('T',tailRange,nu);
117 tailArea = area(tailRange,tailDensity, ...
118      'DisplayName','P-value area');
119 tailArea.FaceColor = 'blue';
120 tailArea.EdgeColor = 'none';
121 hold off
```

```

122 title(['Fig. 4. Theoretical PDF of the t-distribution with shaded ' ...
123       'p-value area',', \nu = ', num2str(nu)]);
124 xlabel('t');
125 ylabel('Density');
126 legend('show');
127
128 % 6.5. Compare the theoretical and simulated p-values
129 theoreticalPValue = 1 - cdf('T',tValue,nu);

```

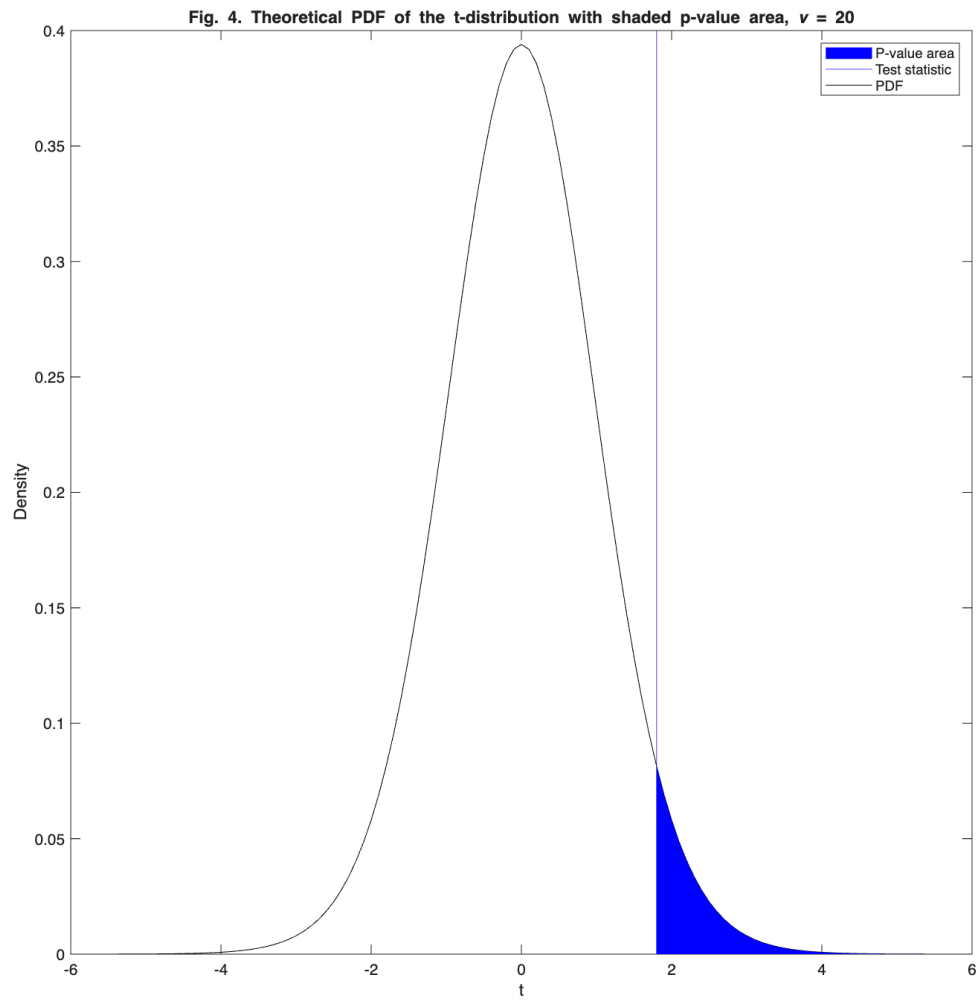


Figure 4: Limiting distribution with values at least as extreme as the observed test statistic shaded

7. Final notes

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